

Team Worksheet — Lab 29

Multivariate SPC and Control Plan Governance (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-29-multivariate-spc-control-plan.xlsx`.

3. What we produced

Deliverable: A Hotelling T-squared chart plus a governed control plan for your capstone.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Understand the false alarm problem: 5 separate charts each at $\alpha = 0.0027$ gives a combined false alarm rate over 1.3 percent — five times intended.

Step 2. Understand the more serious failure: two characteristics can each sit within their own limits while their COMBINATION is impossible, and separate charts will never see it.

Step 3. Identify a set of correlated characteristics in your capstone — dimensions on one part, or related cycle times across linked steps.

Step 4. Compute the correlation matrix and confirm the characteristics are genuinely correlated. If they are independent, separate charts are fine.

Step 5. Learn Hotelling's T-squared: it collapses several correlated characteristics into one statistic accounting for the covariance structure.

Step 6. Build the T-squared chart with its upper control limit from the F-distribution, and note there is no lower limit — T-squared is always positive.

Step 7. Understand the interpretation difficulty: a T-squared signal tells you something is wrong but not which variable. Use decomposition or individual charts as a follow-up diagnostic.

Step 8. Now build your capstone control plan in full: for each critical X and the Y — the metric, target, specification, measurement method, sample size, frequency, chart type, owner and reaction plan.

Step 9. Write the reaction plan for each metric as a specific instruction: containment action, escalation path, named owner and time limit. 'Investigate' is not a reaction plan.

Step 10. Add the control plan to the process documentation and update the SOP, training material and visual management boards.

Step 11. Step up to portfolio governance: as Black Belt you audit control plans across multiple projects. Define your audit cadence and criteria.

Step 12. Define what a control plan audit checks: is the chart still being maintained, are signals being actioned, is the owner still in post, and has the process drifted since handover?

Step 13. Identify the most common decay mode — charts maintained but signals never actioned — and write your countermeasure for it.

7. Self-check

Your correlated characteristics are confirmed by a correlation matrix before applying T-squared, every control plan metric has a specific named reaction plan, and you have defined an audit cadence with explicit criteria.

Complete? ■ Yes ■ Not yet

*Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary
Infotech Academy Pte Ltd*